The Hype Engine: Attention, Herding, & Reversals

Bachelor Thesis Proposal (Behavioral Finance)

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1 Motivation

Behavioral finance emphasizes that investors, especially individual or retail investors, face attention constraints and may trade on salient information rather than fully processing fundamentals, or considering the complete range of investing options they have. A central implication is that attention-driven order flow can generate temporary price pressure, followed by short-horizon reversals once attention fades and liquidity providers absorb imbalances (Barber & Odean, 2008; Barber et al., 2022; Da et al., 2011). This mechanism is closely related to the broader literature on investor sentiment and limited attention, and it offers a disciplined way to interpret short-run return predictability without requiring immediate changes in fundamental cash-flow expectations.

Inspired by this literature, and in particular by the recent work of Barber et al. (2022) on Robinhood herding and the foundational attention-return patterns documented by Da et al. (2011), my thesis proposal aims to investigate the transmission mechanism starting from retail attention to buying behavior and price pressure leading to subsequent reversals in the US equity market. The following sections describe how I plan to address this topic.

2 Research Question & Hypotheses

My thesis aims to investigate the transmission mechanism starting from retail attention to buying behavior and price pressure leading to subsequent reversals in the US equity market. Specifically, I desire to explore the following research question:

> *Do increases in Google-attention Search Volume Index (SVI) and market based attention proxies translate into herding episodes, and does this transmission channel explain the short-run price pressure followed by the medium-run reversal documented in the behavioral finance literature?*

To address this research question methodologically, I will test the hypotheses presented below.

Hypothesis 1. Attention triggers Robinhood demand. Lagged abnormal Google attention spikes predict abnormal increases in Robinhood ownership:

$$\mathbb{E}[\Delta \texttt{RH}_{i,t+1} \mid \texttt{ASVI}_{i,t} \text{ high}] > 0.$$

Mechanism: attention is scarce and search activity is a revealed measure of retail information demand; higher attention increases the likelihood that constrained-attention investors buy the stock (Barber & Odean, 2008; Da et al., 2011; De Giorgi et al., 2026)[1].

[1]For this particular hypothesis, I still need to carefully consider the timing of the attention spikes and the measurement of Robinhood demand, as well as potential confounding factors (e.g., extreme returns, high abnormal volume, earning annoucements) that could simultaneously drive both attention and demand.

In section 2, $\Delta\texttt{RH}_{i,t}$ will measure the change in the percentage of shares outstanding held by Robinhood investors, and $\texttt{ASVI}_{i,t}$, a proxy for attention that leverages Google SVI, will be defined as by Da et al. (2011).

Hypothesis 2. Attention → herding probability. Lagged attention spikes increase the probability and intensity of extreme Robinhood buying ("herding"):

$$\mathbb{P}(\texttt{Herd}_{i,t+1} = 1 \mid \texttt{ASVI}_{i,t} \text{ high}) > \mathbb{P}(\texttt{Herd}_{i,t+1} = 1 \mid \texttt{ASVI}_{i,t} \text{ low}).$$

Empirically, my definition of $\texttt{Herd}_{i,t}$ will be closely aligned with the design of (Barber et al., 2022).

Hypothesis 3. Confirmed attention → price pressure and reversal. Herding episodes generate short-run price pressure followed by medium-run reversal, and these effects are strongest when attention spikes are *confirmed* by Robinhood herding:

$$\mathbb{E}[r^{abn}_{i,t} \mid \texttt{Confirm}_{i,t} = 1] > 0 \quad \text{and} \quad \mathbb{E}[r^{abn}_{i,t+1:t+20} \mid \texttt{Confirm}_{i,t} = 1] < 0.$$

This hypothesis connects the attention–returns patterns documented with Google Trends to the Robinhood herding return patterns by explicitly testing the transmission mechanism search → Robinhood demand and potential herding → returns (Barber et al., 2022; Da et al., 2011, 2025). The precise definition of $\texttt{Confirm}_{i,t}$ is still to be determined, but it will likely be a binary variable.

Hypothesis 4. Small-cap amplification. The attention-to-herding channel and the post-event reversal are stronger among small-cap stocks:

$$\begin{aligned}\mathbb{P}(\texttt{Herd}_{i,t+1} = 1 \mid \texttt{ASVI}_{i,t} \text{ high}, \quad & \texttt{Small}_{i,t} = 1) \\ & > \mathbb{P}(\texttt{Herd}_{i,t+1} = 1 \mid \texttt{ASVI}_{i,t} \text{ high}, \quad \texttt{Small}_{i,t} = 0)\,.\end{aligned}$$

and

$$\begin{aligned}\left|\mathbb{E}\left[r^{abn}_{i,t+1:t+20} \mid \texttt{Confirm}_{i,t} = 1, \quad \texttt{Small}_{i,t} = 1\right]\right| > \Big|\mathbb{E}\Big[r^{abn}_{i,t+1:t+20} \mid \texttt{Confirm}_{i,t} = 1, \\ \texttt{Small}_{i,t} = 0\Big]\Big|\,.\end{aligned}$$

Mechanism: It is plausible that a larger share of the movements in small-cap stocks is more retail-dominated. Furthermore, small-cap stocks are typically less liquid and more volatile. So, attention spikes are more likely to translate into concentrated retail buying and stronger transitory mispricing (Amihud, 2002; Barber & Odean, 2008; Shleifer & Vishny, 1997).

3 Data

3.1 U.S. Stock Returns & Fama-French Factors

For the U.S. stock returns, I will use the CRSP database, which provides comprehensive data on stock prices, returns, and trading volumes for U.S. equities. I will focus on the period from May 2018 to August 2020 at a daily frequency to align with the availability of Robinhood data. I will only consider ordinary common stocks (share codes 10 and 11) listed on the NYSE, AMEX, and NASDAQ exchanges. Finally, I will mitigate the survivorship bias by adjusting the returns for delisting events following the procedure described by Bali et al. (2016).

Additionally, I will compute abnormal returns using the Fama-French five-factor model. This model is widely used in empirically asset pricing research and will allow me to isolate the abnormal return component that may be driven by retail attention and herding behavior. The five factors, as well as the risk-free rate, will be obtained from the Kenneth R. French Data Library[2].

3.2 Robinhood Investors Holdings

This is a key dataset for my thesis and it is fortunately publicly available on the website of Robintrack[3]. Robintrack scraped data from the Robinhood trading platform and provides information on the number of Robinhood users holding each stock for given timestamps, which serves as a proxy for retail demand and potential herding behavior; as depicted by Barber et al. (2022). The data covers the period from May 2018 to August 2020, and was sadly discountinued thereafter by Robinhood. The data frequency is hourly, but I will aggregate it to a daily frequency to match the stock returns data. To proceed, I will likely follow the approach of Barber et al. (2022).

3.3 Google Search Volume Index

To measure retail attention, I will use the Google Search Volume Index (SVI) for each stock ticker. The SVI provides a normalized measure of the search interest for a particular keyword (in this case, the stock ticker) over time. I will obtain the SVI data from Google Trends[4] for the same period as the returns and Robinhood data (May 2018 to August 2020). I will follow the procedure described by Da et al. (2011) to construct a daily abnormal SVI measure that captures attention spikes.

4 Methodology

In this section, I outline the empirical design I intend to use to test the four hypotheses in a way that is defensible and directly comparable to the existing literature on

[2]https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html
[3]https://robintrack.net/
[4]https://trends.google.com/trends/

attention-induced trading and return reversals. The design combines (i) a revealed-attention proxy from Google Trends in the spirit of Da et al. (2011), (ii) platform-level retail demand and herding measures from Robintrack as in Barber et al. (2022), and (iii) standard return and firm-characteristic data from CRSP. The approach follows the canonical structure in this literature: predictive regressions to establish the attention $\rightarrow$ demand channel, and event-study analyses to quantify price pressure and reversal around extreme demand episodes (Barber & Odean, 2008; Barber et al., 2022; Da et al., 2011).

4.1 Sample Construction & Merge

The baseline sample is U.S. common stocks (CRSP share codes 10 and 11 and limited to the NYSE, AMEX, and NASDAQ listed stocks) over the Robintrack coverage window (May 2018 to August 2020), merged by ticker and trading date. CRSP provides returns, prices, volume, and market capitalization; Robintrack provides daily counts of Robinhood holders per ticker; and Google Trends provides search intensity series. Following Barber et al. (2022), I will impose basic filters to reduce mechanical artifacts: (i) require a minimum number of prior-day Robinhood holders (baseline: 100) when defining herding events; (ii) exclude stock-days with missing CRSP returns; and (iii) apply standard microstructure filters such as excluding extremely low prices in robustness checks. I will implement all those filters in a systematic and reproducible way through publicly available code on a dedicated `GitHub` repository[5].

4.2 Key Variables

Retail attention: abnormal search volume (ASVI). Largely inspired by Da et al. (2011) and Barber et al. (2022), retail attention is proxied by abnormal Google search intensity. Let $\texttt{SVI}_{i,t-1}$ denote daily search intensity for ticker i lagged by one day: $t-1$. The baseline abnormalization uses a pre-event baseline window, as presented by Barber et al. (2022).

$$\texttt{ASVI}_{i,t-1} = \ln\left(\frac{\texttt{SVI}_{i,t-1}}{\frac{1}{L}\sum_{\ell=2}^{L+1}\texttt{SVI}_{i,t-\ell}}\right),$$

where L is a lookback window (baseline: $L = 20$ trading days, consistent with the "abnormal SVI" construction used for daily specifications in Barber et al. (2022); robustness will consider longer windows in the spirit of the original weekly ASVI definition in Da et al. (2011)). Attention "spikes" will be defined using within-stock percentiles (e.g. top 5% within stock) and, as robustness, cross-sectional percentiles by day.

[5] https://github.com/YVYX04/Bachelor_Thesis

Robinhood demand and herding. Let $RH_{i,t}$ be the number of Robinhood holders for stock i at time t. Typically, the Robintrack data allow us to observe an hourly frequency. To align with both correlational and causal Robinhood attention work, I will track both absolute and relative changes (Barber et al., 2022; De Giorgi et al., 2026):

$$\Delta RH_{i,t} = RH_{i,t} - RH_{i,t-1}, \qquad UCR_{i,t} = \frac{RH_{i,t}}{RH_{i,t-1}}.$$

Herding events follow Barber et al. (2022): define $\texttt{Herd}_{i,t} = 1$ if stock i is among the top 0.5% of stocks with positive $UCR_{i,t}$ on day t and has at least 100 prior-day holders. This tail-event construction delivers a tractable set of extreme episodes and maps cleanly to the paper's documented return reversals. I aim for consistency with the already existing literature so this seems appropriate, but I will also consider alternative thresholds (top 1%, top 0.2%) and minimum-holder filters as robustness checks.

Confirmed attention. To operationalize Hypothesis 3, I will need to define a binary *confirmed attention* indicator. I propose the following temporary definition, which captures the idea that attention spikes are more likely to generate price pressure and reversal when they are confirmed by contemporaneous herding. However, this might be changed as I explore the data and the timing of attention spikes and herding events.

$$\texttt{Confirm}_{i,t} = \mathbb{1}\{ASVI_{i,t} \geq \tau\} \times \mathbb{1}\{\texttt{Herd}_{i,t} = 1 \text{ or } \texttt{Herd}_{i,t+1} = 1\},$$

where τ is a high attention threshold. This definition allows for a short timing wedge: search activity may precede platform demand by up to one trading day (consistent with the idea that attention leads trading, and consistent with lag structures in the attention literature).

Returns and abnormal returns. Daily returns $r_{i,t}$ come from the daily stocks CRSP file. Abnormal returns $r_{i,t}^{abn}$ will be computed using a market-adjusted benchmark (probably the market return `Mkt-RF` provided by Fama and French) as baseline.

$$r_{i,t}^{abn} = r_{i,t} - \texttt{Mkt-RF}_t.$$

Depending on the results I obtain, I also intend to compute a factor-adjusted abnormal return §$r_{i,t}^{5F}$ using the Fama-French five-factor model, which will allow me to control for systematic risk factors and isolate the abnormal return component more precisely.

$$r_{i,t}^{5F} = r_{i,t} - \texttt{RF}_t - \beta_{MKT}\texttt{MKT}_t - \beta_{SMB}\texttt{SMB}_t - \beta_{HML}\texttt{HML}_t - \beta_{RMW}\texttt{RMW}_t - \beta_{CMA}\texttt{CMA}_t.$$

Following Audrino et al. (2024), the factor loadings β_{MKT}, β_{SMB}, β_{HML}, β_{RMW}, and β_{CMA} will be estimated using a rolling window of 360 days for each stock and a minimum of 200 observations, which allows the factor loadings to adapt over time and capture potential changes in the risk profile of the stocks.

Small-cap classification. To test Hypothesis 4, I will likely define $\texttt{Small}_{i,t}$ using NYSE breakpoints to mimic the approach used by Fama and French (1993). The exact implementation details will be defined later.

4.3 Empirical tests by hypothesis

H1: Attention triggers Robinhood demand

The first test estimates whether attention shocks predict subsequent platform demand. The baseline specification is a panel regression with stock and date fixed effects:

$$\Delta RH_{i,t+1} = \alpha_i + \delta_t + \beta\, ASVI_{i,t} + \gamma' \mathbf{X}_{i,t} + \varepsilon_{i,t+1},$$

where $\mathbf{X}_{i,t}$ includes controls commonly used in the attention and Robinhood herding literature: lagged returns, lagged volatility, abnormal volume, and extreme-return indicators (which proxy salience but also help mitigate confounding by large information events) (Barber & Odean, 2008; Barber et al., 2022; Da et al., 2011). The coefficient of interest is $\beta > 0$.

A key design choice concerns timing and confounds (earnings, news, extreme returns). In the proposal stage, the defensible approach is to (i) use lags, (ii) include date fixed effects, and (iii) include explicit controls that Barber et al. show predict herding (abnormal volume, extreme returns, abnormal news, earnings indicator) as robustness (Barber et al., 2022).

H2: Attention increases herding probability

Next, I test whether attention shocks increase the probability of an extreme herding episode. Following Barber et al. (2022), I estimate a linear probability model (LPM):

$$\texttt{Herd}_{i,t+1} = \alpha_i + \delta_t + \beta\, ASVI_{i,t} + \gamma' \mathbf{X}_{i,t} + u_{i,t+1}.$$

This mirrors Barber et al. (2022) determinants-of-herding setup, where abnormal SVI is included alongside extreme returns and abnormal volume measures. The main parameter is $\beta > 0$.

H3: Confirmed attention generates price pressure and reversal

To quantify price pressure and reversal, I implement an event-study design around $\texttt{Confirm}_{i,t} = 1$ events and compare them to (i) attention-only spikes without herding and (ii) herding events without contemporaneous attention spikes.

Event windows. Consistent with Barber et al. (2022), I intend to define the main post-event horizon as 20 trading days, with additional short windows to capture immediate price pressure:

$$[0,0],\ [1,1],\ [1,5],\ [1,20].$$

However, depending on the results I obtain, it is highly likely that I will also consider a longer post-event horizon (e.g. up to 60 or 120 days) to capture potential longer-term reversals, as well as a pre-event window (e.g. $[-10, -1]$) to check for any anticipatory price movements (in fact, Barber et al. (2022) consider days prior to the herding event). For each event type, I compute average BHARs (and/or CARs) and plot the event-time path from $t-10$ to $t+20$, paralleling the presentation in the Robinhood herding paper.

Overlap rules. I still need to define rules for handling overlapping events, which is a common issue in event studies and is likely to appear with the Robinhood herding events.

Regression-based complement. In addition to event-time averages, I estimate predictive regressions of future abnormal returns:

$$r^{abn}_{i,t+1:t+h} = \alpha_i + \delta_t + \beta\, \texttt{Confirm}_{i,t} + \gamma' \mathbf{X}_{i,t} + \epsilon_{i,t+h}, \qquad h \in \{5, 20, 60, 120\}. \tag{1}$$

The hypothesis predicts $\mathbb{E}[r^{abn}_{i,t} \mid \texttt{Confirm}_{i,t} = 1] > 0$ (pressure) and $\beta < 0$ for $h = 20, 60, 120$ (reversal), consistent with attention-induced trading models and the documented Robinhood herding reversals (Barber et al., 2022; Da et al., 2011).

H4: Small-cap amplification

Hypothesis 4 is tested as a heterogeneity design in both the channel and the return equations. First, the attention-to-herding link:

$$\texttt{Herd}_{i,t+1} = \alpha_i + \delta_t + \beta\, ASVI_{i,t} + \theta\, (ASVI_{i,t} \times \texttt{Small}_{i,t}) + \gamma' \mathbf{X}_{i,t} + \eta_{i,t+1}.$$

Second, the reversal conditional on confirmed attention:

$$r^{abn}_{i,t+1:t+20} = \alpha_i + \delta_t + \beta\, \texttt{Confirm}_{i,t} + \phi\, (\texttt{Confirm}_{i,t} \times \texttt{Small}_{i,t}) + \gamma' \mathbf{X}_{i,t} + \nu_{i,t+20}.$$

The prediction is $\theta > 0$ and $\phi < 0$: attention shocks are more likely to trigger extreme retail herding in small caps, and the subsequent reversal is larger in magnitude. Again, I emphasize that I will likely consider longer reversal horizons (e.g. 60 or 120 days) in robustness checks, as the reversal probably materializes over a longer horizon. This is consistent with the idea that attention-driven retail demand creates stronger price pressure when stocks have less depth and are more retail-dominated (Amihud, 2002; Barber & Odean, 2008; Da et al., 2011; Shleifer & Vishny, 1997).

5 Extension

Differentiating Exogenous Attention Shocks from Endogenous Hype Cascades

It is my understanding that a key empirical challenge in the attention–trading literature is that spikes in search activity can reflect (i) *exogenous* information arrivals (news, earnings, macro shocks) or (ii) *endogenous* feedback dynamics in which retail attention and trading reinforce each other ("hype cascades"). While my baseline design uses lags and controls to mitigate confounding, this extension proposes a direct, data-driven decomposition of attention spikes into exogenous versus endogenous episodes.

Hawkes-process motivation. My approach is inspired by the work of Kang (2026), who models the dynamics of retail demand as a self-exciting point process. Extreme retail-demand episodes are modeled as a self-exciting point process: an unusually large retail surge today raises the near-term likelihood of further surges. A Hawkes process formalizes this intuition through a history-dependent *intensity* (arrival rate) $\lambda(t)$, which increases after past events and decays over time. The central implication is that periods of elevated $\lambda(t)$ correspond to regimes in which retail activity is more *endogenous* (self-reinforcing) rather than driven by new external information.

At this stage, I should warn the reader that my **current mathematical expertise is not sufficient** to fully specify the Hawkes model, so the following steps are a preliminary sketch of the intended approach.

1. Defining Retail Surge Events. Using Robintrack daily aggregates, I define a binary indicator of extreme retail-demand days. For example, let $\Delta RH_{i,t}$ denote the daily change in Robinhood holders for stock i. Define a stock-level surge indicator

$$\mathbb{1}\{\text{Surge}_{i,t} = 1\} = \mathbb{1}\{\Delta RH_{i,t} \geq q_i^{(p)}\},$$

where $q_i^{(p)}$ is the within-stock p-quantile of $\Delta RH_{i,\cdot}$ (e.g. $p = 0.95$). To obtain a market-wide event series, define the daily count of surges

$$C_t = \sum_i \mathbb{1}\{\text{Surge}_{i,t} = 1\},$$

and define a market event when C_t is unusually high:

$$\mathbb{1}\{\mathcal{E}_t = 1\} = \mathbb{1}\{C_t \geq q^{(p)}(C)\}.$$

This delivers event times $\{t_k\}$ that capture clustered episodes of broad retail surging. I can use these event for the Hawkes process.

2. Estimating Retail "Criticality" via a Hawkes Intensity. I fit a (univariate) Hawkes model to the event times $\{t_k\}$ with conditional intensity

$$\lambda(t) = \mu + \sum_{t_k < t} \alpha e^{-\beta(t-t_k)},$$

where μ is a baseline (exogenous) event rate, α captures self-excitation, and β controls the decay of excitation over time. The fitted intensity $\widehat{\lambda}_t$ captures the evolving likelihood of retail surges, and thus serves as a state variable for the degree of endogeneity in retail activity. A standard summary of endogeneity is the *branching ratio* $n = \alpha/\beta$ (Kang, 2026, p. 7):

- If $n < 1$, the process is subcritical: events are mostly exogenous and clusters are small.
- If $n = 1$, the process is critical: events can trigger long cascades and large clusters.
- If $n > 1$, the process is supercritical: events trigger explosive cascades and the process is dominated by endogenous dynamics.

3. Classifying Attention Spikes as Exogenous versus Endogenous. Let $\mathbb{1}\{\text{ASVI_spike}_{i,t} = 1\}$ denote a high-attention event (e.g. top 5% within stock). Then, I would define a high-criticality regime based on the Hawkes intensity:

$$\mathbb{1}\{\text{HighCrit}_t = 1\} = \mathbb{1}\{\widehat{\lambda}_t \geq q^{(0.9)}(\widehat{\lambda})\}.$$

Then classify attention spikes into two types:

$$\begin{aligned} \text{Exogenous attention:} \quad & \text{ASVI_spike}_{i,t} = 1 \ \wedge \ \text{HighCrit}_t = 0, \\ \text{Endogenous hype:} \quad & \text{ASVI_spike}_{i,t} = 1 \ \wedge \ \text{HighCrit}_t = 1. \end{aligned}$$

The economic interpretation is that attention spikes during low $\widehat{\lambda}_t$ are more plausibly information-driven, while attention spikes during high $\widehat{\lambda}_t$ are more consistent with a self-reinforcing retail cascade.

4. Testing Transmission Channel by Type. I re-estimate the core channel and return tests with interactions by the Hawkes state to study whether the attention-to-herding transmission and the subsequent reversal are stronger when attention spikes occur in a high-criticality regime. For herding probability (extension to H2):

$$\texttt{Herd}_{i,t+1} = \alpha_i + \delta_t + \beta_1 \texttt{ASVI}_{i,t} + \beta_2 \text{HighCrit}_t + \beta_3 \big(\texttt{ASVI}_{i,t} \times \text{HighCrit}_t\big) + \gamma' X_{i,t} + u_{i,t+1}.$$

For post-event abnormal returns (extension to H3):

$$r^{abn}_{i,t+1:t+h} = \alpha_i + \delta_t + \theta_1 \texttt{Confirm}_{i,t} + \theta_2 \text{HighCrit}_t + \theta_3 \big(\texttt{Confirm}_{i,t} \times \text{HighCrit}_t\big) + \eta' X_{i,t} + \varepsilon_{i,t+h}.$$

The key predictions are $\beta_3 > 0$ and $\theta_3 < 0$: the attention-to-herding transmission and the subsequent reversal should be materially stronger when attention occurs in a high-criticality (endogenous) retail regime.

WARNING: The Hawkes-process extension is a preliminary sketch and will require significant refinement and validation. I will need to carefully consider the model specification, estimation procedure, and robustness checks to ensure that the classification of attention spikes into exogenous versus endogenous is meaningful and empirically valid. This extension is ambitious (relative to my mathematical expertise) and may be adjusted based on feasibility and data constraints.

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